Project 1: Bug Problem

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1 Part 1

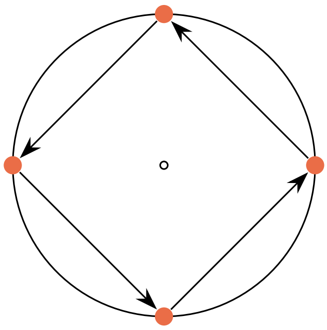

Figure 1: The four bugs are in orange and the arrows point in their initial direction.

2 Part 2

2.1 a)

To analyze the movement of the bugs using calculus, the positions are first defined in Cartesian coordinates. For a bug located at a distance $f(\theta)$ from the origin and an angle θ, the Cartesian coordinates are:

$$(x, y) = (f(\theta)\cos\theta, f(\theta)\sin\theta)$$

For the specific bug released at $(0, a)$, this corresponds to the position where $\theta = \frac{\pi}{2}$ and $f(\theta) = a$.

2.2 b)

The slope of the tangent line between the first bug (at angle θ) and the second bug (at angle $\theta + \frac{\pi}{2}$) describes the direction of the velocity vector. The coordinates of the first bug are $(f(\theta)\cos\theta, f(\theta)\sin\theta)$. Due to the symmetry of the square formation, the second bug is always rotated 90° ($\frac{\pi}{2}$ radians) relative to the first. This means that the coordinates of the target bug are:

$$(f(\theta)\cos(\theta+\frac{\pi}{2}), f(\theta)\sin(\theta+\frac{\pi}{2})) = (-f(\theta)\sin\theta, f(\theta)\cos\theta)$$

The slope m between these two points is:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{f(\theta)\cos\theta - f(\theta)\sin\theta}{-f(\theta)\sin\theta - f(\theta)\cos\theta}$$

Factoring out $f(\theta)$ and -1:

$$m = \frac{f(\theta)(\cos\theta - \sin\theta)}{-f(\theta)(\sin\theta + \cos\theta)} = \frac{-(\cos\theta - \sin\theta)}{\sin\theta + \cos\theta} = \frac{\sin\theta - \cos\theta}{\sin\theta + \cos\theta}$$

2.3 c)

By asserting that the geometric slope found in (b) is equal to the slope of the tangent to the polar curve $r = f(\theta)$, a differential equation is created. The general formula for the slope $\frac{dy}{dx}$ of a polar curve is:

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{f'(\theta)\sin\theta + f(\theta)\cos\theta}{f'(\theta)\cos\theta - f(\theta)\sin\theta}$$

Setting the geometric slope equal to the derivative slope:

$$\frac{\sin\theta - \cos\theta}{\sin\theta + \cos\theta} = \frac{f'(\theta)\sin\theta + f(\theta)\cos\theta}{f'(\theta)\cos\theta - f(\theta)\sin\theta}$$

Cross-multiplying allows us to solve for $f(\theta)$:

$$(\sin\theta - \cos\theta)(f'(\theta)\cos\theta - f(\theta)\sin\theta) = (\sin\theta + \cos\theta)(f'(\theta)\sin\theta + f(\theta)\cos\theta)$$

$$f'(\theta)\sin\theta\cos\theta - f(\theta)\sin^2\theta - f'(\theta)\cos^2\theta + f(\theta)\sin\theta\cos\theta = f'(\theta)\sin^2\theta + f(\theta)\sin\theta\cos\theta + f'(\theta)\sin\theta\cos\theta$$

Canceling common terms ($f'(\theta)\sin\theta\cos\theta$ and $f(\theta)\sin\theta\cos\theta$) from both sides:

$$-f(\theta)\sin^2\theta - f'(\theta)\cos^2\theta = f'(\theta)\sin^2\theta + f(\theta)\cos^2\theta$$

$$-f(\theta)(\sin^2\theta + \cos^2\theta) = f'(\theta)(\sin^2\theta + \cos^2\theta)$$

Using the identity $\sin^2\theta + \cos^2\theta = 1$:

$$-f(\theta) = f'(\theta) \implies f(\theta) = -f'(\theta)$$

The only function of theta that fits this equation is $f(\theta) = Ke^{-\theta}$. Because the known initial distance is a when $\theta = 0$, it must be that $K = a$.

$$f(\theta) = ae^{-\theta}$$

3 Part 3

With $f(\theta)$ known, the arc length formula can be used to find the total length the bug travels.

$$L = \lim_{b\to\infty} \int_0^b \sqrt{(f(\theta))^2 + (f'(\theta))^2}\, d\theta$$

Substituting $f(\theta) = ae^{-\theta}$ and $f'(\theta) = -ae^{-\theta}$:

$$L = \lim_{b\to\infty} \int_0^b \sqrt{(ae^{-\theta})^2 + (-ae^{-\theta})^2}\, d\theta$$

$$L = \lim_{b\to\infty} \int_0^b \sqrt{a^2e^{-2\theta} + a^2e^{-2\theta}}\, d\theta$$

$$L = \lim_{b\to\infty} \int_0^b \sqrt{2a^2e^{-2\theta}}\, d\theta$$

$$L = \lim_{b\to\infty} \int_0^b a\sqrt{2}e^{-\theta}\, d\theta$$

$$L = a\sqrt{2} \lim_{b\to\infty} \left[-e^{-\theta}\right]_0^b$$

$$L = a\sqrt{2} \lim_{b\to\infty} \left(-e^{-b} - (-e^{-0})\right)$$

$$L = a\sqrt{2} \lim_{b\to\infty} \left(1 - \frac{1}{e^b}\right)$$

As $b \to \infty$, the term $\frac{1}{e^b}$ approaches 0. This result tells us:

$$L = a\sqrt{2}(1 - 0) = a\sqrt{2}$$

4 Solving with Series

While the calculus approach guides you through the steps of solving the problem using polar coordinates, I think a much more intuitive solution to this problem can be found using series. You can start by looking at a simpler problem. In the original problem the bug would be constantly changing direction as it chases the next bug. Instead, imagine the bugs move a fraction t of the distance to the next bug before changing direction.

A square can be constructed by drawing the trajectories for each bug. If these trajectories are redrawn each time the bug changes direction a pattern of repeating squares can be seen. Let D be the initial side length of the square formed by the bugs (the distance between two bugs). Using the

Pythagorean theorem, the ratio of the side length of the *next* square (D_{new}) to the current square (D_{old}) after moving a proportion t of the distance is:

$$\text{Ratio } = \frac{D_{new}}{D_{old}} = \sqrt{t^2 + (1-t)^2}$$

The total distance traveled is the sum of these segments. This forms a geometric series:

$$L_{total} = \sum_{n=0}^{\infty} Dt\cdot(\sqrt{t^2+(1-t)^2})^n = Dt + Dt\cdot\sqrt{t^2+(1-t)^2} + Dt\cdot(\sqrt{t^2+(1-t)^2})^2 + \dots$$

The sum of a geometric series is $S = \frac{\text{first term}}{1-\text{ratio}}$. Here, the first term is the distance traveled in the first step, Dt.

$$L_t = \frac{Dt}{1-r(t)} = \frac{Dt}{1-\sqrt{t^2+(1-t)^2}}$$

To find the continuous path of the bug, we take the limit as the step size t approaches 0. This results in the indeterminate form $\frac{0}{0}$, so we multiply by the conjugate to evaluate the limit:

$$L = \lim_{t\to 0} \frac{Dt}{1-\sqrt{t^2+(1-t)^2}} \cdot \frac{1+\sqrt{t^2+(1-t)^2}}{1+\sqrt{t^2+(1-t)^2}}$$

$$L = \lim_{t\to 0} \frac{Dt\left(1+\sqrt{t^2+(1-t)^2}\right)}{1-(t^2+(1-t)^2)}$$

$$L = \lim_{t\to 0} \frac{Dt\left(1+\sqrt{t^2+1-2t+t^2}\right)}{1-(2t^2-2t+1)}$$

$$L = \lim_{t\to 0} \frac{Dt\left(1+\sqrt{2t^2-2t+1}\right)}{2t-2t^2}$$

$$L = \lim_{t\to 0} \frac{Dt\left(1+\sqrt{2t^2-2t+1}\right)}{2t(1-t)}$$

We can now cancel t from the numerator and denominator:

$$L = D\lim_{t\to 0} \frac{1+\sqrt{2t^2-2t+1}}{2(1-t)}$$

Evaluating the limit by substituting $t = 0$:

$$L = D\left(\frac{1+\sqrt{0-0+1}}{2(1-0)}\right) = D\left(\frac{1+1}{2}\right) = D$$

From this limit it can be seen that the total distance the bug travels is exactly equal to the initial distance between the bugs, D. In our specific problem, the bugs start at coordinates like $(a, 0)$ and $(0, a)$. The distance between them is the hypotenuse of a right triangle with legs a:

$$D = \sqrt{a^2 + a^2} = a\sqrt{2}$$

This result tells us the total length traveled is $a\sqrt{2}$.